

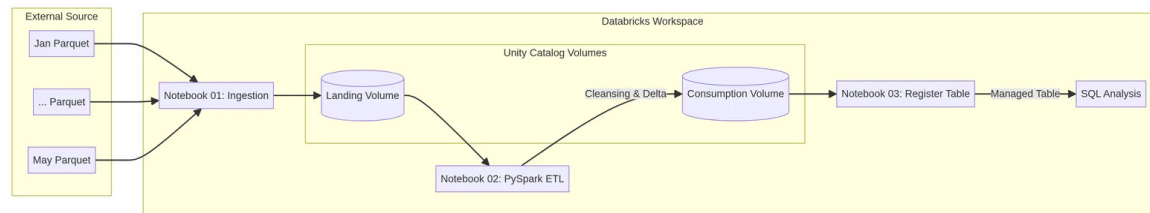


# NYC YELLOW TAXI DATA LAKE ON DATABRICKS

Technical Case Study: Data Architect Challenge

DATA ENGINEERING EXCELLENCE • 2023 STUDY

# MODERN MEDALLION ARCHITECTURE ORCHESTRATES DATA FLOW



## NOTEBOOK 01: INGESTION

Moves raw monthly Parquet files from external sources into the Unity Catalog Landing Volume.

## NOTEBOOK 02: TRANSFORMATION

PySpark engine applies cleansing rules and converts data to high-performance Delta format.

## NOTEBOOK 03: REGISTRATION

Registers Delta files as Managed Tables in Unity Catalog for SQL access.

## ANALYTICS LAYER

Final business queries executed directly on registered tables for actionable insights.

# ROBUST FOUNDATION FOR ENTERPRISE-SCALE DATA ENGINEERING

## Architecture Summary

Successfully delivered a governed, multi-layered Data Lake using modern Databricks features on a restricted tier, proving scalability and security.

## Strategic Implementation

Leveraged Medallion architecture principles to ensure data quality, from raw ingestion to high-performance analytical consumption.

**READY FOR PRODUCTION DEPLOYMENT**

## GITHUB REPOSITORY STRUCTURE

### /src

01\_ingestion.ipynb

02\_transformation.ipynb

03\_registration.ipynb

### /analysis

queries.sql

/

README.md

requirements.txt

architecture\_diagram.pdf